

B. Pharm. 8th Semester, Pharmaceutical Biotechnology (BP 6021)

2nd sessional Exam, April 2025

S =
N =
P =

Time: 1 hr 30 minutes

Max Marks: 30

Instructions: - Part A - All questions are compulsory

Part B - Attempt any two questions

Part C - Attempt any one question

Part A

- (i) Nucleosome
- (ii) Basic structural components of an immunoglobulin molecule.
- (iii) Principle behind the ELISA technique?
- (iv) Type of molecule is detected by Southern blotting and by Western blotting.
- (v) Point mutation and give one possible consequence it can have on a protein.

(5x2 marks)

Part B

- (i) What are the two main classes of MHC molecules, and how do they differ in structure and function?
- (ii) Name one key difference between humoral and cell-mediated immune responses.
- (iii) Explain the structure of immunoglobulins, highlighting the role of variable and constant regions in antigen binding and effector functions.

(2x5 marks)

Part C

- (i) Describe the ELISA technique in detail. Mention its types, steps involved, and applications in disease diagnosis.
- (ii) Define mutation and explain different types of mutations. Discuss their impact on protein function and possible implications in genetic diseases.

(1x10 marks)